

Team: Keyboard Gremlins

COS301 Capstone Project
University of Pretoria

Hackathon Platform

Admin Portal & Participant Portal User Guide



User Manual

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1. Introduction

Welcome to the Hackathon Platform

The Hackathon Platform is a cloud-agnostic system purpose-built for running deterministic optimization hackathons. Participants generate an output from a set of provided inputs, rules, constraints, and levels, and an official solver grades that output to produce a score and the highest score wins. The platform gives organizers a highly configurable way to run events, and give participants a single place to register, form teams, submit solutions, and track their standing on a live leaderboard.

System Overview

Running an optimization hackathon manually is operationally heavy as someone has to distribute problem files, collect submissions, run the solver, and keep a leaderboard up to date, usually under time pressure as a deadline approaches. This platform standardizes that entire lifecycle:

Event setup → Registration → Team management → Submissions → Scoring → Leaderboard → Insights

The platform is split into two portals:

- Admin Portal: used by event organizers to configure events, manage problems and levels, run live operations during the event, and review insights.
- Participant Portal: used by hackathon participants to register for events, form or join teams, download problem resources, submit solutions, and track the leaderboard.

Document Purpose

This manual is your complete guide to using the Hackathon Platform, covering both portals. Whether you're the organizer setting up an event or a participant submitting a solution, this manual explains what each screen does and how to use it.

How to Use This Manual

- New Users: start with “Getting Started” and “Account Management”
- Admins: go to Part A - Admin Portal
- Participants: go to Part B - Participant Portal
- Part C : covers features shared by both portals, plus the Help and Support

2. Getting Started

What is the Hackathon Platform?

The hackathon Platform is a web application with two portals- an Admin Portal and a Participant Portal - built around a single core idea: organizers publish a problem, participants submit outputs against it, and an official solver scores every submission automatically and consistently.

Key Benefits

- Standardized event lifecycle: the same setup -> submissions -> scoring -> leaderboard flow works for any optimization challenge
- Consistent, automated scoring: every submission is graded by the same official solver, removing manual grading inconsistency
- Live visibility: participants see live leaderboard standing and a countdown timer; organizers see live event statistics
- Configurable events: public or private events, adjustable team sizes, dynamic levels, and controlled re-scoring after solver fixes

3. User Roles and Responsibilities

The hackathon Platform supports two primary user types, each with its own portal and permissions:

Admin(Event Organizer)

- Primary Role: creating, configuring, and operating hackathon events

Key Responsibilities:

- Create and configure events (name, schedule, team size, rules, visibility)
- Upload and maintain the problem statement, levels, resources, and solver
- Monitor registrations, team submissions, and platform health during the event
- Run live operations: extend time, freeze/unfreeze the leaderboard, push solver fixes
- Review statistics and insights to support decisions during and after the event

Participant

Key Responsibilities:

- Register for an event (and provide the event password/key for private events)
- Create a team or request to join an existing one
- Download and review the problem statement, rules, and level/resource files
- Submit solution outputs and source code archives for each level
- Review scoring feedback and logs, and track the live leaderboard

4. Account Management

4.1 Creating an Account

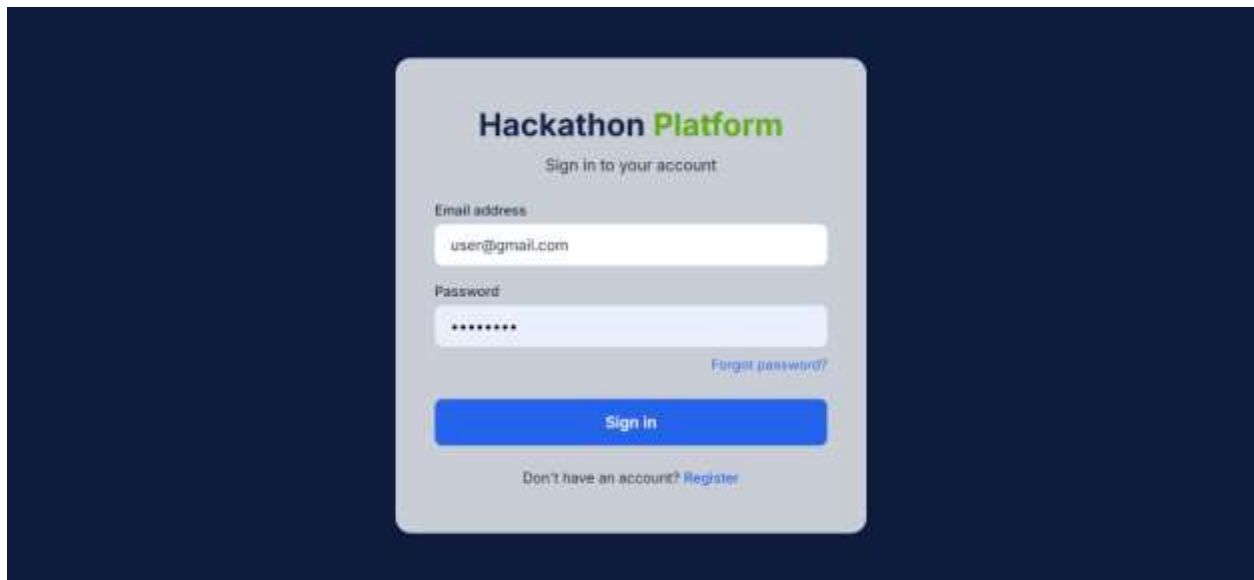
A screenshot of a web form titled "Hackathon Platform" with the subtitle "Create a new account". The form is set against a dark blue background. It contains several input fields: "First name" and "Last name" (two separate white boxes), "Email address" (a single white box containing the placeholder "you@example.com"), "Password" and "Confirm Password" (two separate white boxes with masked characters). Below these fields is a prominent blue button labeled "Create account". At the bottom of the form, there is a link that says "Already have an account? Sign in".

Both Admins and Participants use the same account system, with access to each portal controlled by the account's assigned role.

1. Navigate to the platform's landing page and select "Login"
2. Complete the registration form:
 - Username: must be unique across the platform
 - Email Address: used as your login identifier
 - Password: choose a strong password

3. Submit the form. Your account is created and ready to use immediately

4.2 Logging In

A screenshot of the 'Hackathon Platform' login page. The page has a dark blue background. In the center is a light gray rounded rectangle containing the login form. At the top of this rectangle is the text 'Hackathon Platform' in white, with 'Platform' in green. Below it is the text 'Sign in to your account' in a smaller white font. The form has two input fields: 'Email address' with the value 'user@gmail.com' and 'Password' with masked characters '*****'. To the right of the password field is a link 'Forgot password?'. Below the input fields is a blue 'Sign in' button. At the bottom of the form is a link 'Don't have an account? Register'.

4. Navigate to the login page.
5. Enter your registered email and password.
6. Select "Login." You'll land on the Admin Dashboard or the Participant Dashboard depending on your account's role.

4.3 Profile Management

From the account menus you can view your account details, update notification preferences, and change your password.

4.4 Logging Out

Use the “Sign Out” option in the account menu to safely end your session, especially on a shared computer.

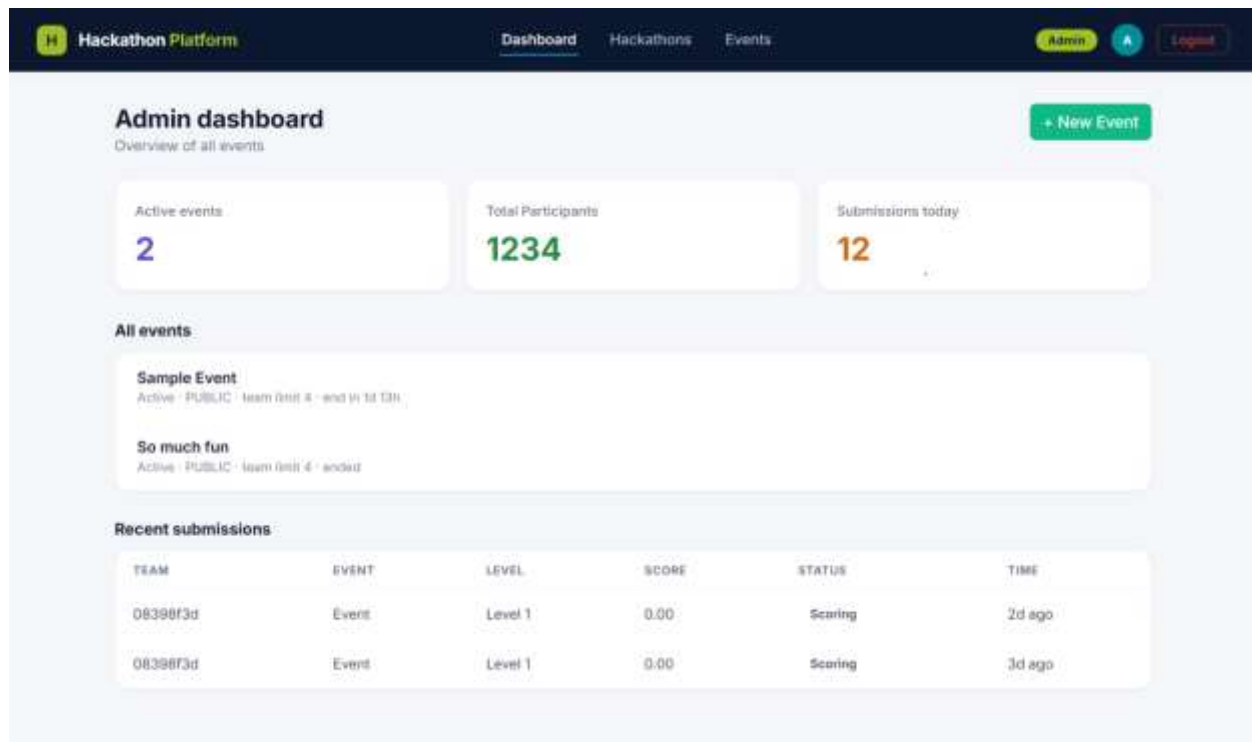
PART A

Admin Portal

5. Admin Portal Overview

The Admin Portal is organized into three layers. The Admin Dashboard gives you a snapshot across everything you run. Below that, Hackathons is the top-level list of the hackathons you organize (e.g “Entelect Challenge 2026”). Each Hackathon can contain one or more Events, and each Event has its own management area with tabs for Overview & Live Ops, Levels & Files, Solver, Registrations, Teams, Submissions, and Analytics.

5.1 Admin Dashboard



The Admin Dashboard is the landing screen after logging in as an Admin, giving you an overview of all events.

- **+ New Event** - button in the top-right corner that takes you straight to the Create Event form

Stats at a glance

- Active Events - count of events currently running
- Total Participants - total participants across all events
- Submissions Today - how many submissions have come in today

All Events

A compact list of your events, showing each event's name, its number of teams, and the

time remaining until it ends. If you haven't created any events yet, this shows "No events created yet."

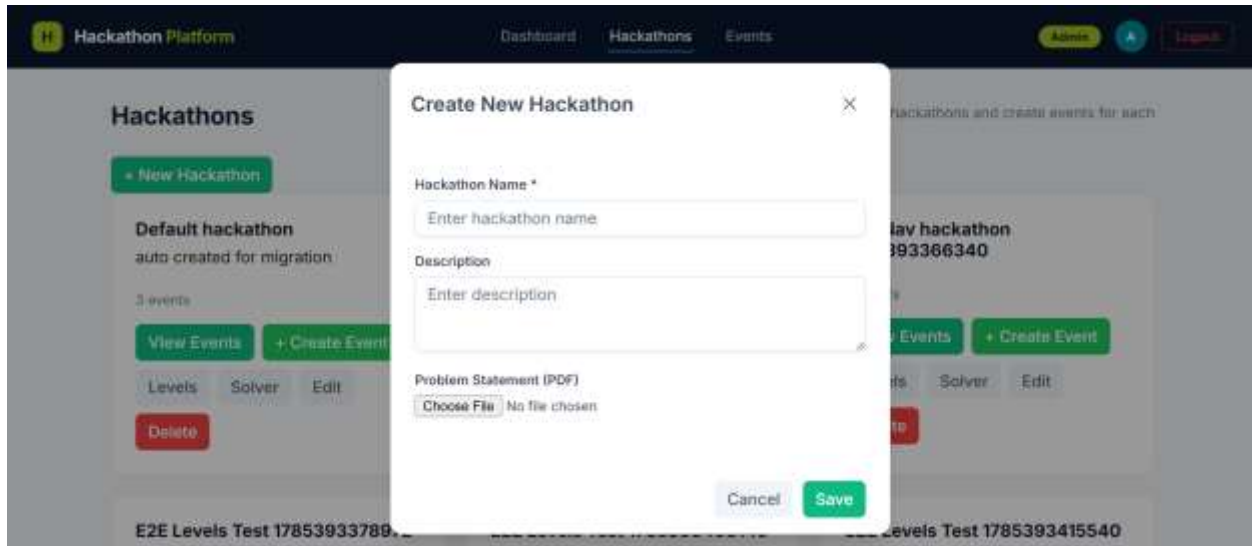
Recent Submissions

A table of the latest submissions across all your events, with columns for Team, Event, Level, Score, Status (Shown as a coloured badge), and Time. If nothing has been Submitted yet, this shows "No submissions yet."

5.2 Managing Hackathons

The Hackathons screen lists every hackathon you organize as a card, and is the starting point for creating events, managing details, and drilling into levels or the solver.

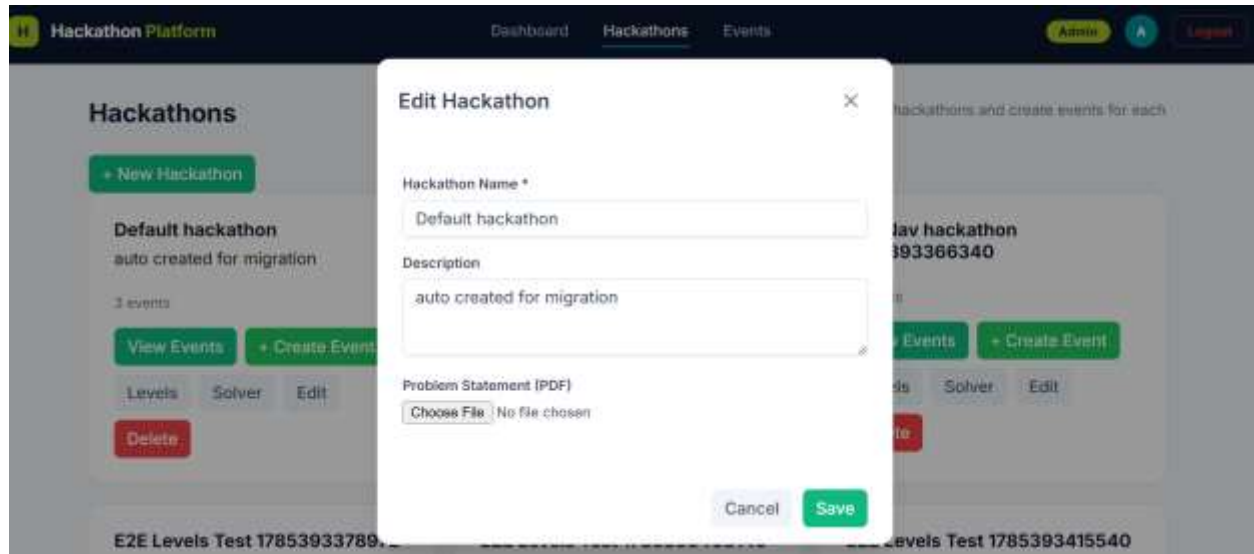
1. Creating a Hackathon



7. Select “+ New Hackathon.” This opens the Create/Edit Hackathon dialog.
8. Enter the Hackathon Name (required)
9. Add a Description
10. Select “Save.” A confirmation toast appears in the top-right corner, and the new hackathon appears as a card.

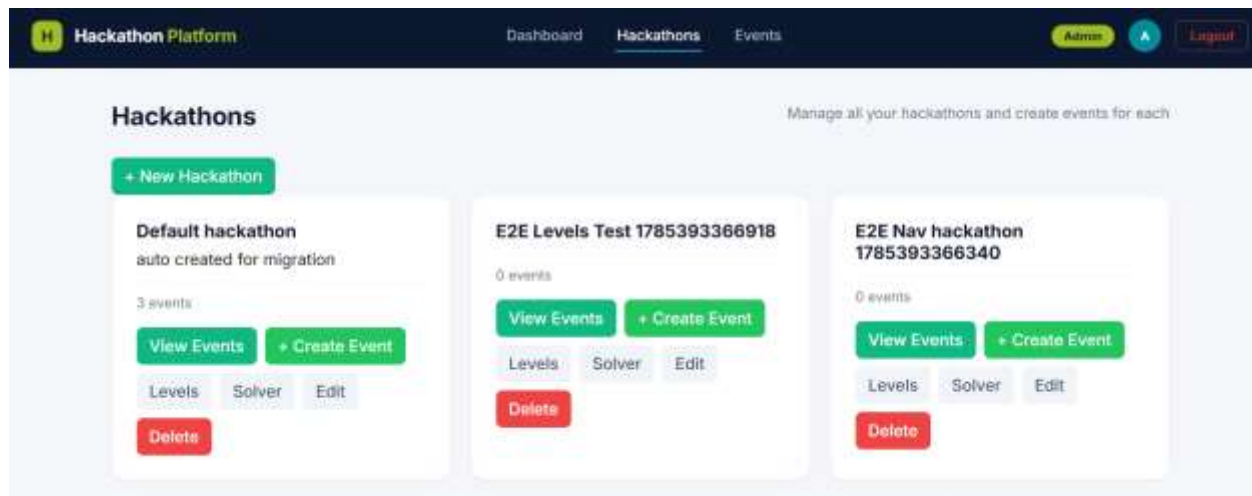
Select “Cancel” at any point to close the dialog without saving.

2. Editing a Hackathon



Select “Edit” on a hackathon’s card to reopen the same dialog, pre-filled with its current name and description, so you can update them.

3. Hackathon Card Actions



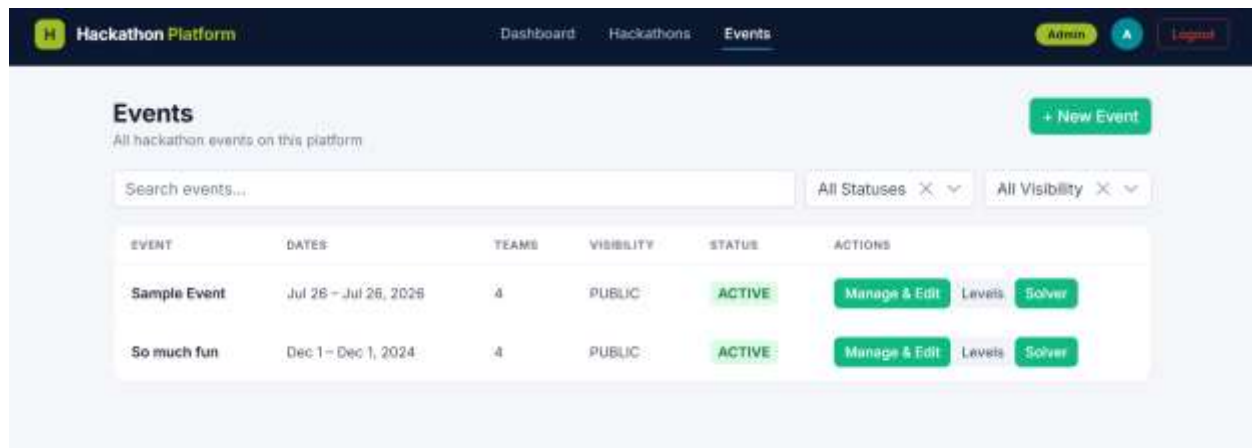
Each hackathon card shows its name, a status badge, its description, and how many events it contains along with these actions:

- View Event: Opens the Events list, filtered to this hackathon
- +Create Event: opens the Create Event form, scoped to this hackathon
- Manage: opens the Manage screen to edit the hackathon's core details
- Levels: jumps directly to Levels & Files for this hackathon
- Solver: jumps directly to Solver Management for this hackathon
- Edit: opens the Create/Edit dialog to rename or update the description
- Delete: permanently removes the hackathon

If you haven't created any hackathons yet, the page shows "No hackathons created yet" with a shortcut to create your first one.

5.3. Managing Events

1. Viewing & Filtering Events



The Events screen lists every event on the platform(or, when reached via “View Events” on a hackathon card, the events under that hackathon - in which case a back button is shown at the top). From here you can:

- Search events by name using the search box
- Filter by status using the status dropdown
- Select “+New Event” to create another event

The events table shows, for each event: Event(name), Dates(start-end), Teams(the event’s configured team size limit), Visibility, and Status (shown as a coloured tag). Each row has three actions:

- Manage & Edit: opens the Manage screen for that event
- Levels: opens Levels & Files for that event
- Solver: opens Solver Management for that event

While events are loading, the page shows “Loading events...”; if no events match your search or filters, it shows “No events match your filter.”

2. Creating a New Event

The screenshot shows the 'Create new Event' page in the Hackathon Platform. The page has a dark blue header with the logo 'H Hackathon Platform' on the left and navigation links 'Dashboard', 'Hackathons', and 'Events' in the center. On the right of the header are buttons for 'Admin', a user profile icon, and 'Logout'.

Below the header, there is a 'Back to Events' button. The main heading is 'Create new Event' with a subtext 'Configure all event parameters before publishing.' The form is divided into three sections:

- Basic Information :**
 - Event Name**: A text input field.
 - Start Date & Time ***: A date and time picker showing '2024/12/01 09:00'.
 - Duration (hours) ***: A text input field showing '48'.
 - Team Size Limit ***: A text input field showing '4'.
- Visibility & Access**: Two radio button options.
 - Public**: 'Anyone can view and register for the event.' (This option is selected).
 - Private**: 'Only invited users can view and register for the event.'
- Rules & Description**: A large text area for entering rules and description.

At the bottom right of the form are two buttons: 'Save Draft' and 'Create Event'.

Select “+ New Event” from the Dashboard, the Hackathons page, or the Events list to open the Create Event Form. Select “Back to Events” at any time to return without finishing.

Basic Information

- Event Name - required
- Start Date & Time - required
- Duration(hours) - required, minimum1
- Team Size Limit - required, minimum1

Visibility & Access

Choose one of the two toggle options:

- **Public:** anyone can view and register for the event
- **Private:** only invited users can view and register; selecting this reveals a required Registration Key field, which participants must enter to register

Rules & Description

A free-text area for the event’s rules and description.

Submitting

Select “Save Draft” to save your progress without publishing, or “Create Event” to finish. Both buttons show a loading state (“Saving...”/“Creating...”) and are disabled while the request is in progress. If something goes wrong, an error message appears at the top of the form.

3. Managing an Event

The Manage screen (“Manage Hackathon”) is where you edit an event’s core details and upload its resources. A back button and a “Back to Hackathons” button both return you to the previous list; “Delete Hackathon” permanently removes it. While details are loading, the page shows “Loading event details...”, an error or success message appears above the form as you make changes.

Basic Information

- Event name - required
- Start date & time - required
- End date & time - required
- Description - the event’s rules and description

Visibility & Access

The same Public/ Private toggle as event creation - Public (“Anyone can register”) or Private (“Registration key required”).

Hackathon Status

A status dropdown lets you set the event’s lifecycle status (for example, upcoming or active). Select “Update Status Only” to save just the status without resubmitting the rest of the form - useful for a quick state change during a live event.

Hackathon Resources

Upload the Problem Statement as a PDF: click the upload area or drag a file onto it, then select “Upload to Azure.” The button stays disabled until a file is selected. A success or error message appears once the upload completes. Uploaded resources become available to registered participants once the event starts.

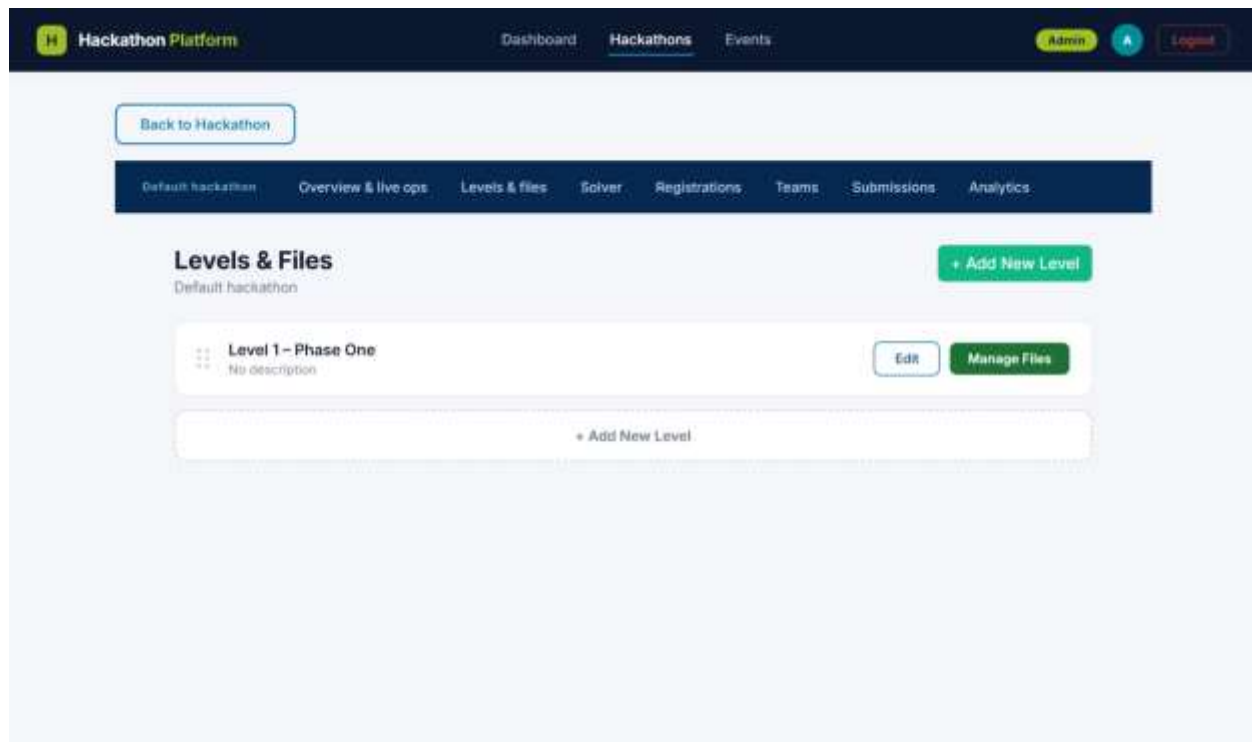
Saving

Select “Cancel” to discard changes and go back, or “Save Changes” to save.

5.4. Levels & Files

Levels & Files, Solver, and the other event-management pages share a sub-navigation bar showing the hackathon’s name and row of tabs: Overview & Live Ops, Level & Files, Solver, Registrations, Teams, Submissions, and Analytics. A “Back to Hackathon” button sits above it. Use the tabs to move between an event’s management pages without losing your place.

1. Adding and Editing Levels



The Levels & Files page lists every level for the event as a card, in order. Select “+ Add New Level” (in the page header, or as a row at the bottom of the list) to add one. If no levels exist yet, the page shows “No levels created yet.”

Each level card shows a drag handle, the level’s name and difficulty, and two actions:

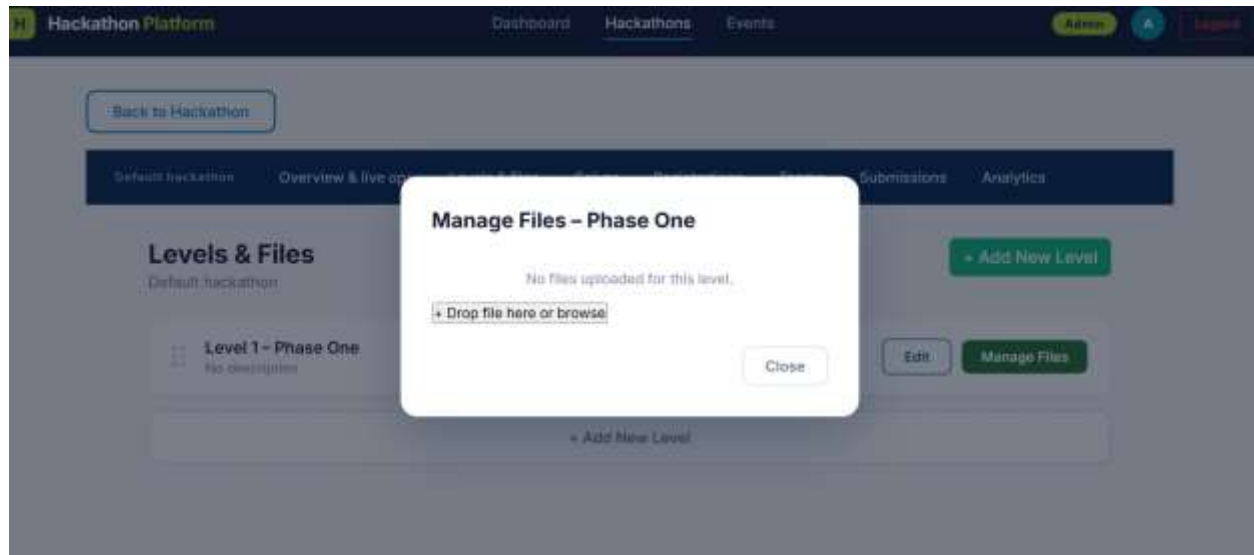
- Edit: opens the level in the Add/Edit Level dialog
- Manage Files: opens the Manage Files dialog for that level

The Add/Edit Level dialog captures:

- Level Name - for example, “Level 1 - Introduction”
- Difficulty - Introduction, Intermediate, Advanced, or Expert
- Scoring Mode - Highest score wins, Lowest time wins, or Time

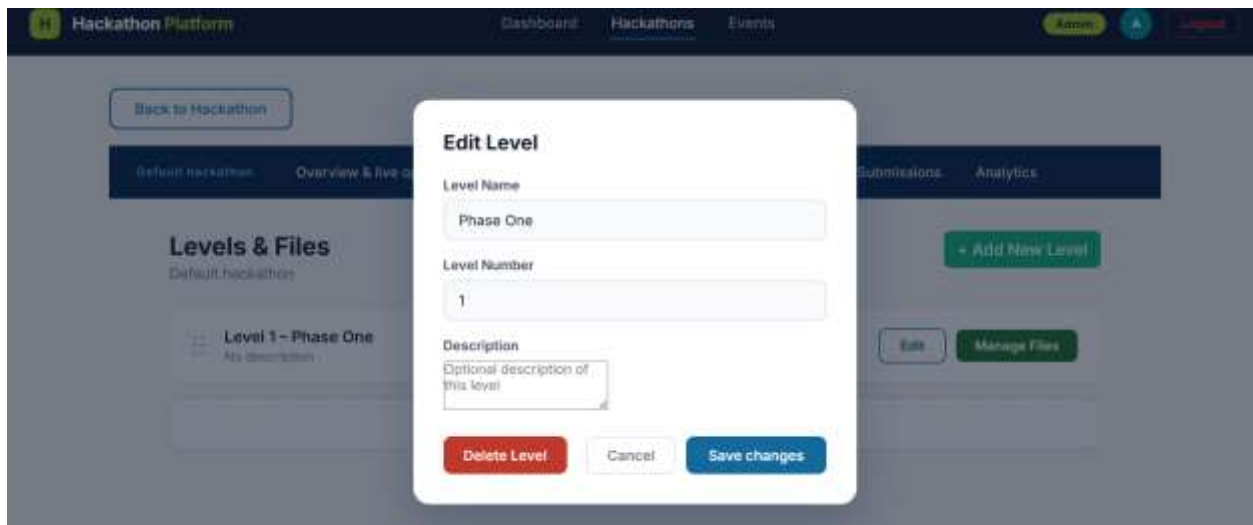
When editing an existing level, a “Delete Level” button is also available. Select “Cancel” to close without saving, or “Add Level”/ “Save changes” to confirm.

2. Managing Level Files



The Manage Files dialog lists the files currently uploaded for a level, each with a “Remove” button, or “No Files uploaded for this level” if none exist yet. Click the drop zone or drag files onto it to upload - multiple files can be added at once. Select “Close” when you’re done.

5.5 Solver Management



The Solver Management page is where you upload and manage the official solver that grades all participant submissions. The solver is a Python script that takes a participant's output file, the level input files, and the level number, and returns a JSON result with the score and validation messages

The Solver Management page shares the same event sub-navigation and “Back to Hackathon” button as Levels & Files. Select “Upload new version” in the header to jump straight to the upload form.

1. Uploading a New Solver Version

The upload card accepts a solver file in `.py` format via drag-and-drop or by browsing.

Solver Requirements:

- The solver must be a Python script (`.py` file)
- The script must accept exactly 3 command-line arguments in the following order:
 1. `output_file` - Path to the participant's submission output file
 2. `level_inputs_dir` - Path to the directory containing the level's input files
 3. `level_number` - The numeric identifier of the level being scored

Solver management

Entelect Challenge 2026

Rescore all submissions

Upload new version

Upload solver version

Solver file (.py)

+ Drop solver file here or browse

Change notes

What changed in this version?

Upload & activate

Version History

VERSION	UPLOADED	RUNS	AVG RUNTIME	ERROR RATE	STATUS
No solver versions uploaded yet.					

Solver Output Format:

The solver must output a single line of valid JSON as its final line of output. The JSON must follow this schema:

```
{  
  
  "score": 85.5,  
  
  "status": "SCORED",  
  
  "errorType": null,  
  
  "messages": [  
  
    "Validation passed for all constraints",  
  
    "Output format is valid"  
  
  ]  
}
```

JSON Fields:

- **score** (number, required) - The numeric score for this submission. Higher scores indicate better performance. Must be a decimal number (e.g., 85.5).
- **status** (string, required) - Must be either **"SCORED"** or **"FAILED"**. If the solver encounters any validation errors or issues, return **"FAILED"**.

- **errorType** (string, optional) - If status is "FAILED", this should indicate the type of error (e.g., "MALFORMED_OUTPUT", "RUNTIME_ERROR", "INVALID_FORMAT"). Can be **null** for successful runs.
- **messages** (array of strings, optional) - A list of human-readable messages providing feedback to the participant. Can be empty for successful runs.

Solver Environment:

- The solver runs in a temporary isolated directory
- The script is executed with the Python interpreter configured by the platform

Upload Process:

1. Alongside the file, provide:
 - **Change Notes** - A short description of what changed in this version (optional)
2. Select "Upload & Activate" to upload the file and make it the active solver used for scoring

Important Notes:

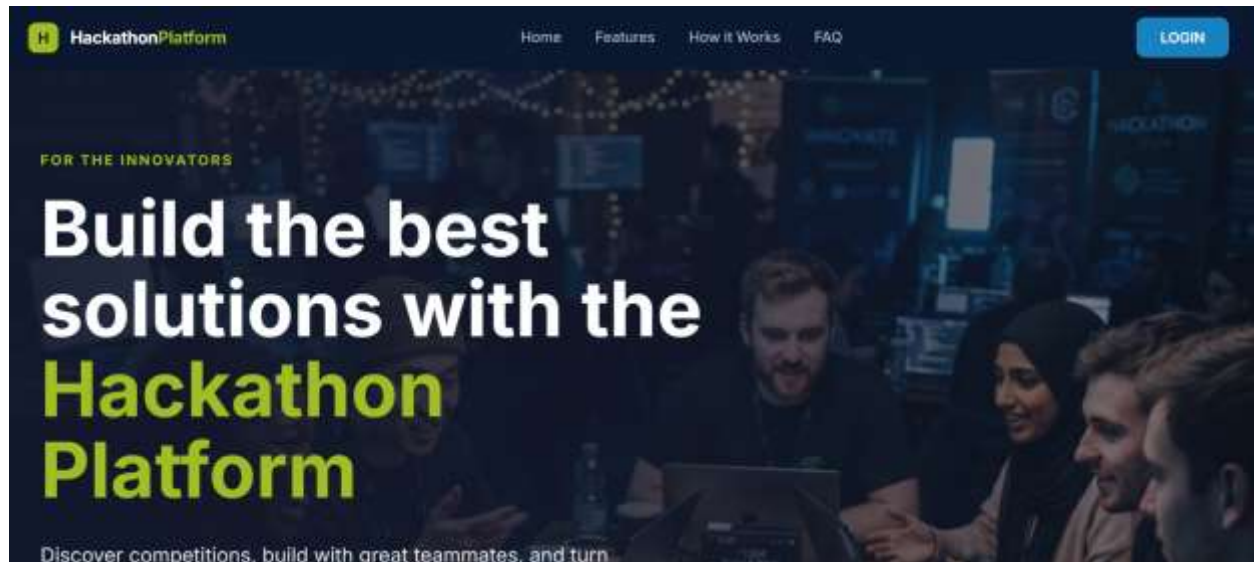
- Only one solver version can be active per hackathon at a time
- Uploading a new solver version automatically deactivates all previous versions
- All submissions submitted before a solver update will retain their original scores
- To re-score submissions with the new solver, use the admin rescore functionality in the scoring operations

2. Version History

Below the upload card, a table lists every solver version that's been uploaded, with columns for Version, Uploaded(date), Runs, Av Runtime, Error Rate, and Status (Shown as a badge, for example "Active"). If no versions have been uploaded yet, this shows "No solver versions uploaded yet." **(NOT INTEGRATED FOR DEMO 2)**

Participant Portal

6. Participant Portal Overview

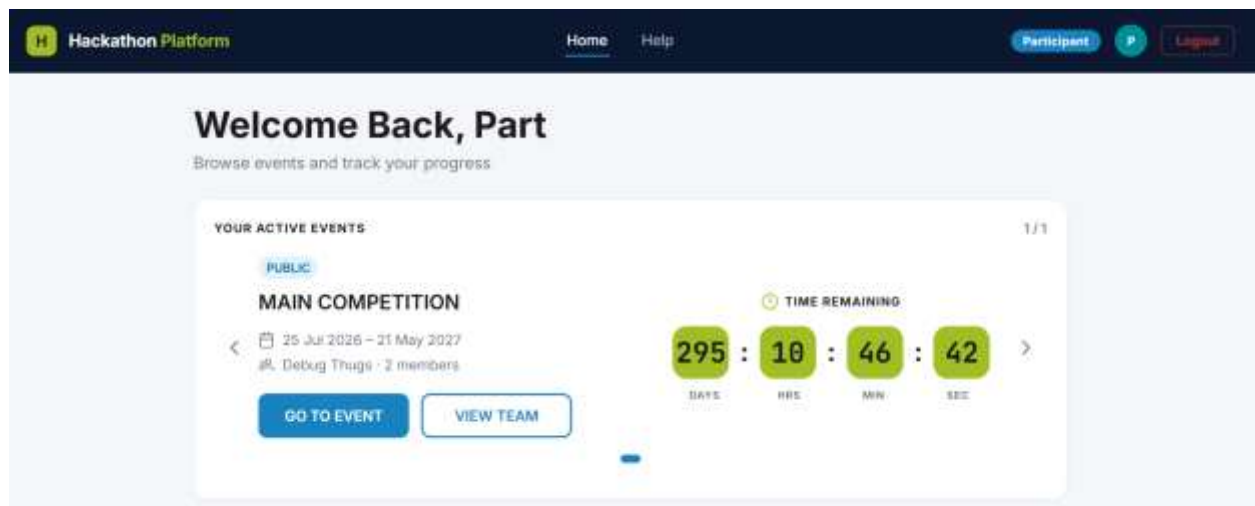


Before logging in, participants land on the public homepage, with a top navigation bar linking to Home, Features, How it Works, and FAQ, plus a Login button. Once logged in, participants land on the Home Dashboard, which surfaces any events you're actively taking part in and a searchable list of open events to register for. Opening a specific event brings up that event's workspace - its own tab bar with Overview, Rules, Upload Solution, Submission History, My Team, and Leaderboard.

6.1 Home Dashboard

The Home Dashboard greets you with “Welcome back, [your name]” and the tagline “Browse events and track your progress.” If an error occurs with loading your data, an error message appears at the top of the page.

1. Your Active Events



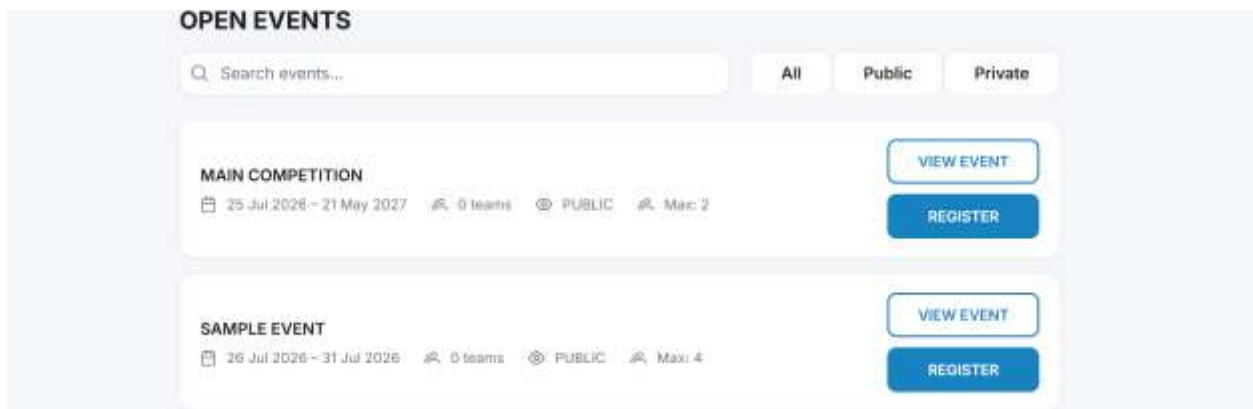
If you're registered for at least one active event, a “Your Active Events” card shows a carousel - with a counter (for example, “1/3”) if you have more than one - that you can page through. Each slide shows :

- A Public or Private tag for the event
- The event's name
- Its dates, and your team's name and the event's team size limit
- **Go to Event** - opens the event's workspace
- **View Team** - opens the My Team tab directly

Each slide also shows a live countdown timer with a label and a

days : hours : minutes : seconds display, counting down to the event's next milestone.

2. Browsing & Registering for Open Events

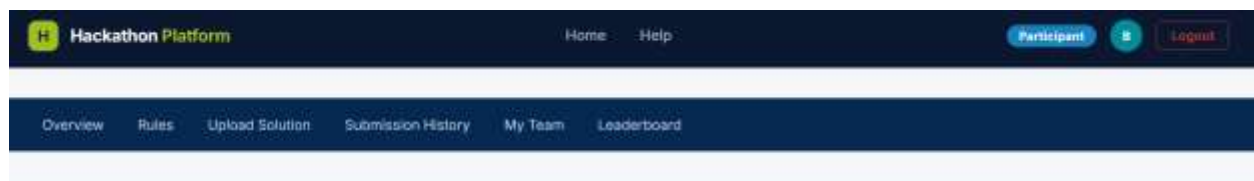


Below your active events, the “Open events” section lists events you can join, with a search box (“Search events....”) and a filter control above the list. While events are loading, the page shows “Loading....”; if there are none available, it shows “No open events are available right now ”; if your search has no matches, it shows “No events match your search.”

Each open event row shows its name, dates, how many teams have already registered, its visibility (with a lock icon for Private or an eye icon for Public), and the event's maximum team size. Two actions are available:

- **View Event:** opens a preview of the event's workspace
- **Register:** registers you for the event and starts your team for it - for a private event, you'll be asked for the registration key set by the organizer.

6.3 The Event Workspace

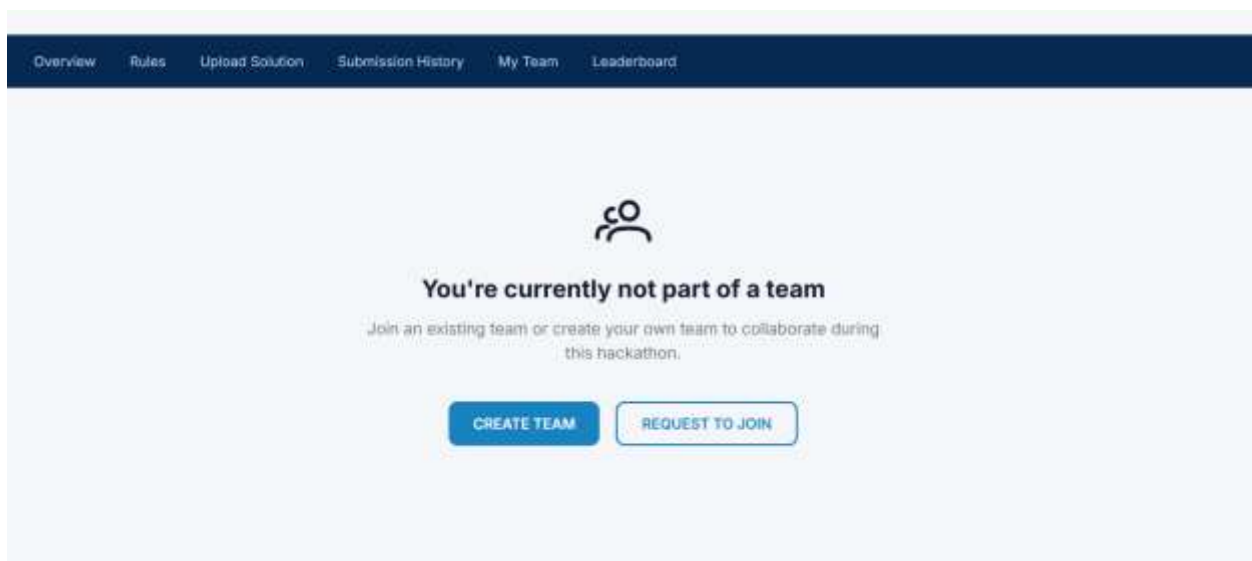


Once you open an event - via "Go to Event" or "View Event" - you land in that event's workspace, with a tab bar across the top:

- **Overview** - the event's summary
- **Rules** - the event's rules and description
- **Upload Solution** - submit your solution output and source code
- **Submission History** - your past submissions, scores, and logs
- **My Team** - create, join, or view your team for the event
- **Leaderboard** - live standings for the event

Each tab loads its own content within the event workspace, so you can move between them without losing your place.

6.4 Team Creation & Management



Open the MyTeam tab within an event's workspace - or select "View Team" on your active event card on the Home Dashboard - to manage your team for that event.

1. Creating a Team

If you haven't joined a team yet, you can create a new team for the event. You'll typically be set as the team's initial member/owner.

2. Joining a Team

Alternatively, you can request to join an existing team for the event. Joining follows an approval flow - an existing team member (or even team owner) needs to approve your request before you're added.

6.5 Accessing the Problem & Resources

Within an event's workspace, the Overview tab gives you the event's summary, and the Rules tab gives you the full rules and description the organizer configured for the event. Once the event starts, any resource files the organizer has uploaded - such as the problem statement - become available here as well.

6.6 Making a Submission

The screenshot shows the 'SUBMIT SOLUTION' page on the Hackathon Platform. The header includes the platform name, navigation links (Home, Help), and user status (Participant, Logout). The main navigation bar lists: Overview, Rules, Upload Solution, Submission History, My Team, and Leaderboard. The page title is 'SUBMIT SOLUTION' with a subtitle: 'Select a level below to view its requirements and submit your solution.' A note states: 'Join or create a team for this event before submitting a solution.' The 'Phase One' section is active, featuring a '1 PHASE ONE' indicator and a 'DOWNLOAD LEVEL FILES' button. Two upload boxes are present: 'UPLOAD SOURCE CODE ARCHIVE' for ZIP files and 'UPLOAD SOLUTION OUTPUT FILE' for JSON files. Each box contains a cloud upload icon, instructions to drag and drop or click to browse, and a respective 'CHOOSE' button. A 'SUBMIT SOLUTION' button is located at the bottom center.

Hackathon Platform Home Help Participant Logout

Overview Rules Upload Solution Submission History My Team Leaderboard

SUBMIT SOLUTION

Select a level below to view its requirements and submit your solution.

Join or create a team for this event before submitting a solution.

Phase One

1 PHASE ONE

DOWNLOAD LEVEL FILES

UPLOAD SOURCE CODE ARCHIVE



Drag and drop your ZIP here
or click to browse files

CHOOSE ZIP FILE

UPLOAD SOLUTION OUTPUT FILE



Drag and drop your JSON here
or click to browse files

CHOOSE JSON FILE

SUBMIT SOLUTION

1. Submission Requirements

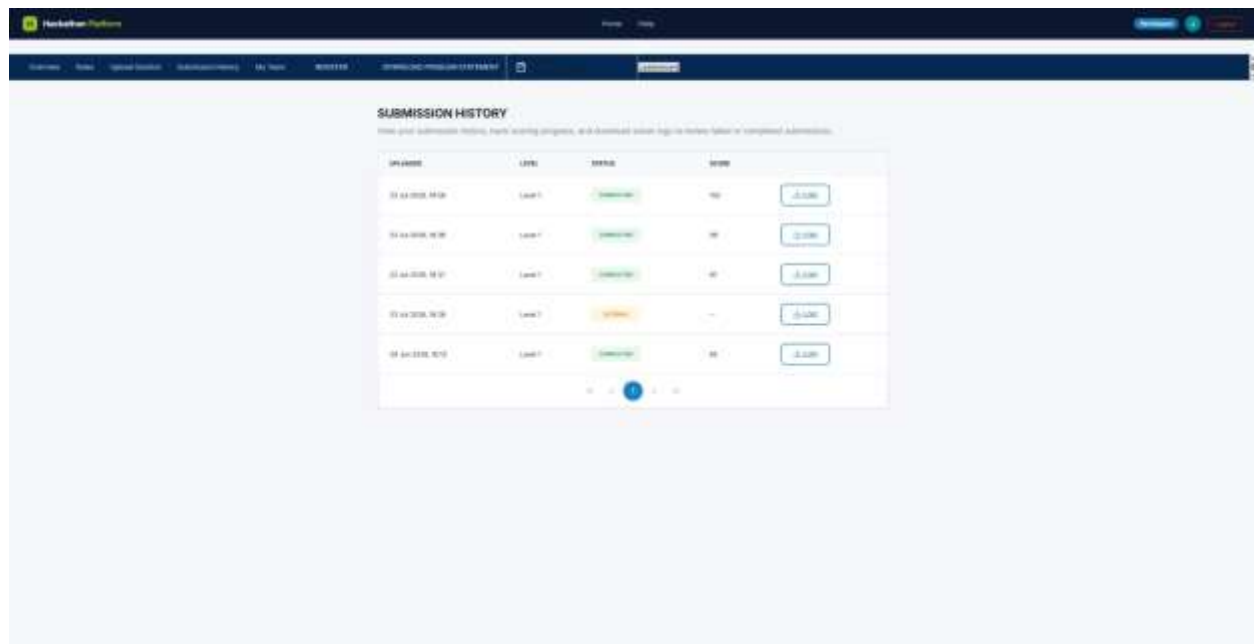
For each level, your team must upload two things:

- Solution Output File: the artifact that the official solver grades
- Source Code Archive(.zip): a zipped archive of the implementation that generated the output

2. Submission Process

11. Open the event and select the “Upload Solution” tab.
12. Upload your solution output file.
13. Upload your zipped source archive.
14. Submit before the event deadline. The platform queues your submission for scoring by the official solver.

6.7 Scoring Feedback & Logs



Open the “Submission History” tab within an event to see your team’s submissions. For each one, you’ll see:

- Your score for that submission
- A success/error status
- Detailed logs and messages - for example, malformed output, rule violations, or validation failures

Your full submission history is listed per level, so you can review or download the logs for any past attempt.

6.8 Leaderboard & Timer



H Hackathon Platform

HomeHelp

ParticipantPLogout

OverviewRulesUpload SolutionSubmission HistoryMy TeamLeaderboard

CURRENT STANDINGS

See how your team ranks against the competition.

#3

team3 — Your Team

0.00 pts

TOP TEAMS

The three highest scoring teams in the competition.

team2

0.00 pts

RANK #2

Debug Thugs

85.00 pts

RANK #1

team3

0.00 pts

RANK #3

FULL RANKINGS

RANK	TEAM	SCORE
1	Debug Thugs	85.00
2	team2	0.00
3	team3	0.00

<<

<

1

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>>

Open the “Leaderboard” tab within an event to see live standings. A countdown timer - also shown on your active event card on the Home Dashboard - displays the timer remaining as days, hours, minutes, and seconds.

PART C

Authentication & Security

Both portals share the same underlying platform service:

- Role-Based Access Control: your account’s role (Admin or Participant) determines which portal and features you can access
- Secure Sessions: the platform uses JWT-based authentication with secure session handling
- Content & File Storage: problem statements, level input files, solver versions, participant outputs, logs, branding assets, and source code archives are all stored and served securely by the platform.

PART D

Help and Support



[Documentation](#)

User Guide

The complete user guide for organizers and participants.

[Open User Guide](#)[FAQ](#)

Frequently Asked Questions

Quick answers to the questions participants ask most often.

How do I join an event?



How do I create or join a team?



How do I submit my solution?



Can I submit multiple times?



How are submissions graded?



Where can I view my submission history?



The in-platform Help Centre opens with the heading “How can we help you?” and the description “Welcome to the help menu, find answers to frequently asked questions.” It offers two ways to find answers: the complete User Guide, and a set of frequently asked questions.

1. User Guide

Under Documentation, select “Open User Guide” to open the complete user guide for organisers and participants - this manual.

2. Frequently Asked Questions

The FAQ section - described as “Quick answers to the questions participants ask most often” - is an expandable accordion covering:

- **How do I join an event?**
Browse the open events and select “Register” before the registration deadline; once accepted, the event appears in your Active Events Carousel
- **How do I create or join a team?**
Open the Teams section inside your event using the “View Team” button, then create a new team or join an existing one
- **How do I submit my solution?**
Select “Go to Event,” open the Upload Solution tab, upload your source archive and solution output, then submit before the event deadline
- **Can I submit multiple times?**
Yes- you can upload new submissions until the submission deadline; your latest valid submission for each level is used for the final results

- **How are submissions graded?**

The platform automatically evaluates your submissions using the organizer's solver and generates a score along with a feedback log

- **Where can I view my submission history?**

Open your event and go to the Submission History tab to view your past submissions

